ROBUSTFOLIO · AN OPEN-SOURCE PORTFOLIO BENCHMARK

The best portfolio in my study lost.

I couldn’t tell you whether that was true.

I built an open-source benchmark: seven ways to build a portfolio, 18 US ETFs, 168 months, after trading costs, one rule fixed before the run. The rule said no. So I checked whether it could have said yes.



The test’s 80% detection threshold, in net excess Sharpe: return above cash per unit of risk, after trading costs. An improvement at the bar would usually be missed.

What this is

A benchmark for portfolio methods: seven constructions, one decision rule, and a test suite that checks the evaluation itself. Not a trading strategy, and not investment advice.

Why it matters

A model can be built correctly and still be judged by a test that cannot see its effect. That is the failure this project exists to catch — and the first one it caught was mine.

The result is a correction of the test, not a win for the portfolio.

THE PROBLEM

Portfolio construction is an estimation problem.

The standard method estimates average returns and correlations from history, then picks the portfolio with the best trade-off between return and risk. The estimates are noisy. The optimizer can load on assets that looked good by chance. With 18 assets and 60 months of history, there are more parameters than independent observations; regularization — shrinking noisy estimates — is the standard response.

The project asks a narrow question: do "robust" methods — a shrinkage covariance (which pulls noisy correlations toward a simpler estimate), a Black–Litterman mean (which blends the historical average with a prior) — beat a plain baseline out of sample, after costs, once you account for having tried more than one method?

Why it matters

A better estimator is only useful if it survives the noise, the costs, and the decision. That is what the benchmark tests.

This is a question about evidence, not a pitch for a strategy.

WHAT I BUILT

A benchmark, not a backtest.

The engine	A walk-forward test: re-estimate each month on the previous 60 months, then test the next month. 168 out-of-sample months in 14 non-overlapping folds.
The universe	18 US ETFs across equity, bonds, commodities and real estate; eligibility rules applied at each date.
The methods	Five standard baselines plus two variants that change one assumption in the standard mean-variance model.
The apparatus	A pre-registered decision rule, a leakage test suite, a numerical check that the optimizer solved the problem, and a reproduction check.

1176 optimizer solves; every one checked numerically. The code and the frozen data snapshot are public.

The benchmark is the deliverable, not just the headline number.

HOW I JUDGED IT

One metric, one rule, fixed before the run.

Net excess Sharpe: return above cash, after 10 bps (0.10%) one-way trading costs, divided by volatility. A summary of the path, not a forecast.

The rule: a method had to beat the best baseline by at least 0.20 net Sharpe, clear the significance tests (the edge is unlikely to be luck), and pass five more checks on turnover, drawdown, cost levels, folds and concentration.

Why both tests

Significance alone admits tiny effects. Size alone admits noise. Requiring both, plus robustness, is what makes a null informative.

The rule was written before the run. The run is exploratory because the planned blind test was compromised.

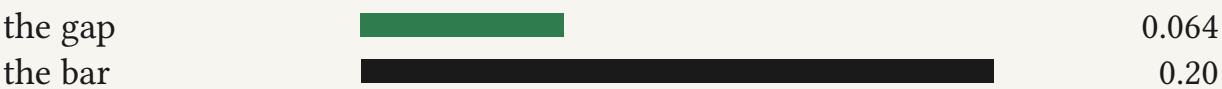
THE RESULT

It finished first. It still failed.

0.8538

net excess Sharpe — the highest of the seven

Gap over the best baseline	+0.064
Materiality bar	0.20
Original test (probability edge is real)	0.587 of 0.95
Checks failed	3 of 8



A ranking and a verdict are different objects.

The number is a summary of one path, not a return you could spend.

The bar was 0.20. The test needed 0.72.

A test has a detection threshold: the smallest true effect it can reliably notice. This one needed about 0.72 to have an 80% chance of noticing it. The bar was 0.20. A real 0.20 improvement would have been missed most of the time.



What went wrong

The original test compared the candidate's Sharpe with the baseline's observed Sharpe. The design was built for a paired comparison. The test was measuring a different question.

A test can be rigorous and still answer the wrong question.

THE CORRECTION

The same data, the right test.

0.917

probability M2 beats B3 on a paired resampling test

Sharpe gap	+0.064
90% range	−0.011 to 0.147
p-value	0.083
Original test	0.587

The two strategies hold almost the same portfolio ($\rho = 0.986$), so comparing them month by month is far more sensitive than comparing each to zero.

The original test was not wrong. It was answering a different question.

THE READING

The data exclude a 0.20 advantage.

0.147 < 0.20

the 90% upper bound is below the materiality bar

NOT MATERIAL

M2 is plausibly a little better. It is not materially better — not enough to justify the extra complexity. The data exclude the 0.20 improvement the study was designed to detect.

This is a bounded result: material, not material, or undecided. Here it is not material.

The result is a bound, not a shrug.

The cap did not rescue shrinkage.

I thought the 30% position cap was doing the job of keeping the portfolio simple, which would explain why the shrinkage method added nothing. I tested it.

Position limit	Shrinkage	Black–Litterman
0.20	−0.015 (−0.107, 0.073)	+0.049 (−0.031, 0.140)
0.30	−0.004 (−0.096, 0.080)	+0.064 (−0.011, 0.147)
0.40	+0.010 (−0.084, 0.097)	+0.119 (0.037, 0.206)
1.00	−0.003 (−0.095, 0.083)	+0.138 (0.044, 0.237)

Shrinkage: not supported. The gap stays at or below zero; every interval includes zero.

Black–Litterman: depends on the cap. The gap grows as the cap loosens, but remains below 0.20. The looser cap also means larger positions: the largest holding rises from 0.30 to 0.80.

The mechanism test did not support the shrinkage explanation. It did show the cap binding the mean treatment.

I ran the new sample. It also failed.

18 country ETFs · 2005–2026 · 252 months · 21 folds. The best baseline was 1/N, not sample MVO.

Method	Net Sharpe	Turnover
B1	0.3527	0.83
B2	0.3423	0.86
M2	0.3193	2.48
B5	0.2919	1.40
B3	0.2797	2.54
B4	0.2715	1.76
M1	0.2710	2.43

M2 vs B1: −0.033 [−0.125, 0.075]. **M1 vs B1:** −0.082 [−0.175, 0.029]. Both upper bounds are below 0.20 both won 0 of 5 cost levels; both traded about 3.0× the baseline.

What replicated — and what did not

The benchmark lesson replicated: the best baseline changed from B3 to 1/N. The treatment advantage did not.

The treatment edge was universe-conditional. The benchmark lesson was not.

WHAT THIS SHOWS

The benchmark is the binding constraint.

The first sample: `M2` beat `B3`, but not the 0.20 bar.

The new sample: equal weight beat both treatments.

The correction: a better test on the same data, not a new strategy.

As an ML student, the lesson I would carry: a model can be well implemented and still be judged by a test that cannot see its effect.

What I would check in my own work

Before trusting a metric, ask what decision it is supposed to support. A test can be rigorous and still answer the wrong question.

Code, paper and the reproduction check: github.com/kira-ml/robustfolio